

Steam single-player pricing analysis — detailed findings

This document expands on the results in `steam-price-analysis.ipynb`. It is written for review before presenting to stakeholders. Numbers below were produced from the cleaned dataset (`data/single-player-games-cleaned.parquet`, **10,736 games**) using the same train/test split and feature sets as the notebook (`random_state=42`, 80/20 holdout).

1. Business problem

Indie and mid-tier publishers need evidence-informed **list prices** on Steam. The core question is:

What list price and game characteristics are **associated** with stronger estimated sales on Steam?

We proxy “sales strength” with **SteamSpy owner-range midpoints** (`owners_mid`, modeled as `log_owners_mid`). That is a noisy, cross-sectional estimate—not verified unit sales. All conclusions below are about **patterns in historical catalog data**, not causal pricing rules.

2. Data and methodology

2.1 Source and scope

Item	Detail
Input	<code>data/single-player-games.csv</code> (~10.7k paid, single-player Steam games)
After cleaning	10,736 rows in <code>single-player-games-cleaned.parquet</code>
Target	<code>log_owners_mid</code> = log of SteamSpy owner-range midpoint
Price field	<code>price_usd</code> (list price at snapshot time)
Features	<code>price_usd</code> , <code>age_days</code> , tag/developer/publisher counts, top-18 genre flags (<code>genre__*</code>), top-15 tag flags (<code>tag__*</code>)

2.2 Cleaning filters (sequential)

Games were dropped if they lacked usable owner estimates, price, release date, or genre/tag metadata needed for modeling. All retained games have **paid** list prices (minimum observed **\$0.49**; no free-to-play rows in the final set).

2.3 Important limitations

- **SteamSpy estimates** are ranges, not audited sales; many games sit at the **10,000-owner floor** (median `owners_mid` = **10,000**).
- **Single snapshot** — no price history, sales events, or wishlist data.
- **Engagement fields** (`steamspy_ccu` , median playtime) are often zero in this extract and were not primary predictors.
- **Cross-sectional association $\neq$ causation** — successful titles may charge more *because* they are hits, not the other way around.

3. Exploratory data analysis (§8–§8b)

3.1 Price distribution

Statistic	Value
Median list price	\$5.99
75th percentile	\$12.99
Mean list price	\$9.30
Maximum	\$199.99

Most games cluster in the **\$3–\$13** band. The long right tail (premium and niche titles) pulls the mean above the median.

Chart: `visualizations/price_usd_hist.png`

3.2 Owner / sales proxy distribution

Estimated owners are **highly skewed**. On a log scale the distribution is more usable for regression. Many games share the same lower bound (10k owners), which compresses variation at the low end.

Statistic	<code>owners_mid</code>
Median	10,000
75th percentile	35,000

Statistic	owners_mid
Mean	142,912 (inflated by blockbuster outliers)
Max	35,000,000

Chart: visualizations/owners_mid_log_hist.png

3.3 Price vs. estimated owners

Pearson correlation between price_usd and log_owners_mid: ≈ 0.29 .

That is a **modest positive** relationship: higher-priced games tend to have somewhat higher estimated owners, but the scatter is wide. Genre, quality, marketing, and franchise effects dominate what this simple correlation can explain.

Charts: visualizations/price_vs_log_owners_scatter.png ,
visualizations/discount_vs_log_owners_scatter.png

3.4 Genre patterns

Genre (primary)	Share of catalog
Action	4,901 games (~46%)
Adventure	2,223
Casual	1,609

Mean log owners by genre flag (multi-hot) shows **Strategy**, **Simulation**, and **RPG** titles with slightly higher typical estimated reach than Action, but differences are moderate and overlap is large.

Charts: visualizations/price_by_primary_genre_box.png ,
visualizations/mean_log_owners_by_primary_genre_bar.png

3.5 Tag patterns (§8b)

Among the top 15 Steam tags encoded as features, tags associated with **higher mean log owners** (among games carrying that tag, $n > 50$) include:

Tag	Mean log owners	Games with tag
Great Soundtrack	11.08	1,222
Atmospheric	11.00	1,580
Story Rich	10.99	1,399

Tag	Mean log owners	Games with tag
Singleplayer	10.88	3,901

These tags correlate with **premium, narrative, or polish-heavy** experiences—not necessarily with “charge more.” They help explain why metadata beyond price improves prediction.

Charts: `visualizations/tag_prevalence_top15.png`,
`visualizations/mean_log_owners_by_top_tags.png`

3.6 Numeric feature correlations

Age (`age_days`, median $\approx$ **3,186 days** $\approx$ 8.7 years) and counts (developers, publishers, tags) show weak to moderate relationships with owners. No single numeric feature besides price stands out as a dominant driver.

Chart: `visualizations/correlation_heatmap_numeric.png`

4. Linear regression and “optimal price” (§9)

4.1 Model specification

OLS predicts `log_owners_mid` from:

- `price_usd`
- `age_days`, `tag_count`, `developer_count`, `publisher_count`
- Top-K **genre__*** flags (no tag flags in this section—narrower, interpretable spec)

Train/test split: 80/20, `random_state=42`.

4.2 Results

Metric	Value
Test R^2	$\approx$ 0.30
Price coefficient	+0.023 (positive on log owners)

A positive price coefficient means that, holding genre and counts fixed, the fitted line associates **higher list price with higher estimated owners**. That is almost certainly **confounding**: hit games can sustain higher prices *and* accumulate more owners. It does **not** imply that raising price causes more sales.

4.3 Revenue proxy curve

The notebook sweeps a synthetic **reference game** (median counts, modal Action genre) across a price grid and plots **revenue proxy** = `price × exp(predicted log owners)`.

Because the price term enters positively, the curve often peaks at the **top of the grid (~\$40)**—the highest price tested—not at the catalog median (~\$6). Treat that peak as a **model artifact**, not a launch-price recommendation.

Charts:

- `visualizations/optimal_price_revenue_curve.png` — exploratory revenue proxy vs. price
- `visualizations/ols_actual_vs_predicted_test.png` — holdout fit quality
- `visualizations/ols_residuals_test.png` — systematic errors (heavy tails, underestimation of blockbusters)

4.4 Stakeholder takeaway for §9

Use this linear model for **transparent what-if conversations** ("if we price like a typical Action game at \$X, what does the formula predict?"). Do **not** use the revenue-curve maximum as a pricing target.

5. Model comparison (§10)

Seven regressors were compared on the **full feature set** (price, counts, all `genre__*` and `tag__*` columns). Same 80/20 holdout and `random_state=42`.

5.1 Holdout performance (test set)

Model	Test R ²	5-fold CV R ² (train)	RMSE	MAE
Gradient Boosting	0.439	0.416	1.059	0.783
Random Forest	0.434	0.406	1.064	0.780
MLP	0.351	0.330	1.138	0.849
KNN	0.332	0.322	1.155	0.829
OLS	0.331	0.308	1.156	0.867
Ridge	0.330	0.308	1.157	0.868
Lasso	0.329	0.308	1.158	0.867

Primary metric: test R² (share of variance in log owners explained on holdout data). **RMSE/MAE** measure typical error on the log scale (lower is better).

5.2 Interpretation

1. **Tree ensembles win** — Gradient Boosting and Random Forest improve test R^2 by roughly **+0.10–0.11** over linear models (~ 0.44 vs. ~ 0.33). Nonlinear **interactions** among price, genre, and tags matter for this proxy target.
2. **Linear models are tied** — OLS, Ridge, and Lasso perform similarly once genre + tag dummies are included; regularization does not materially change rankings here.
3. **MLP is middle tier** — Beats linear models slightly but trails trees; may need more tuning/data to justify complexity.
4. **KNN $\approx$ linear baseline** — Local similarity in feature space adds little beyond global linear structure for this feature matrix.
5. **CV vs. test** — CV R^2 tracks test R^2 without large gaps, suggesting modest overfitting for tree models at the chosen hyperparameters.

Even the best model explains **less than half** of the variance in log owners. A large share of success is **unobserved** (quality, marketing, timing, franchise, visibility).

Charts:

- `visualizations/model_comparison_test_r2.png`
- `visualizations/model_comparison_rmse_mae.png`
- `visualizations/model_comparison_train_vs_test_r2.png`
- `visualizations/model_comparison_cv_vs_test_r2.png`
- `visualizations/best_model_actual_vs_predicted.png`
- `visualizations/best_model_feature_importance.png`

5.3 Feature importance (best tree model)

The winning model's importance plot highlights which metadata fields the ensemble uses most. Typically **price**, select **genre/tag** flags, and **age** rank highly; exact ordering should be read from the regenerated plot after running §10. Importance reflects **predictive contribution**, not causal effect.

6. Synthesis: what we can and cannot claim

What the analysis supports

- **Typical Steam list price** for this segment is about **\$6**; most games sit below **\$13**.
- **Price alone** is a weak signal; **genre, tags, and unmeasured quality** dominate.
- **Nonlinear models** predict the ownership proxy more accurately than linear baselines when rich metadata is available.
- **Transparent linear models** are still useful for scenario framing, with clear caveats about confounding.

What the analysis does not support

- "Set your price to \$X and maximize revenue."
 - "Higher price causes more owners."
 - "SteamSpy midpoint equals true sales."
 - Causal **price elasticity** from a single cross-section.
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7. Recommendations

For product / publishing stakeholders

1. **Anchor on market comps** — Same genre, tag profile, and price band as your title; median catalog price (~\$6) is a starting point, not a formula output.
2. **Use models for discussion, not decree** — Compare scenarios ("at \$9.99 vs. \$14.99, what does the model expect for similar metadata?") while naming assumptions.
3. **Validate before launch** — Wishlists, demos, playtests, and regional pricing experiments outweigh retrospective catalog regression.
4. **Match model to question** — Use **§9 OLS** when explaining trade-offs; use **§10 tree model** when prioritizing forecast accuracy on similar metadata.

For analytics / data team

1. Refit on **title-specific comp sets** rather than the full catalog.
 2. Add **review scores**, **Metacritic**, or **Steam review %** if available.
 3. Build a **longitudinal panel** (price changes, sales events) for elasticity.
 4. Replace or blend SteamSpy with **first-party sales** where possible.
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8. Next steps

Priority	Action
High	Curate a comp universe per upcoming title and refit
High	Collect wishlist / demo conversion data for launch titles
Medium	Ingest review sentiment and update features
Medium	Track price history and promo windows for causal work
Low	Hyperparameter tuning for MLP / deeper ensembles (diminishing returns vs. data quality)

9. Reproducing results

1. Follow setup in [README.md](#) .
2. Run all cells in [steam-price-analysis.ipynb](#) .
3. Generated artifacts:
4. `data/single-player-games-cleaned.parquet`
5. `visualizations/*.png`

Section numbers in this document map to notebook sections **§8–§11**. If you change `TAG_TOP_K`, genre counts, or filters, re-run the notebook and refresh the metrics above.

10. Quick reference — executive summary

Topic	Headline
Dataset	~ 10,736 paid single-player games
Typical price	Median \$5.99
Price vs. owners	Correlation ~ 0.29 ; weak alone
§9 linear “optimum”	Often top of price grid → confounding, not advice
Best predictors	Gradient Boosting / Random Forest , test R^2 ~ 0.44
Linear baseline	Test R^2 ~ 0.33 with full genre + tags
Bottom line	Use for exploration and comps , not causal launch pricing